

CANDY DIALOGUE ENGINE

Operators

1.1.1

This documents lists all operators supported by the `calculate_variable_value()` function in `Candy_Engine.gd`. These operators are used by the `$Set` and `$Flag` commands, but the function itself could be used by your own code for other purposes.

Note that some operators work differently depending on whether a value is provided. For example, if a value is provided, `'strip_edges'` removes trailing spaces at the start and end of the value, and assigns it to the variable; if no value is provided, `'strip_edges'` removes trailing spaces on the variable's current value.

Operators may not always work the way you'd expect in GDScript. Please read this document to see what to expect, and test your dialogues to ensure that operator behavior conforms to your expectations.

Additional operators can easily be added, including operators/methods that don't exist in GDScript. Feel free to request more if you need any.

Basic

`"=`

Assign the value directly to the variable.

`"+=`

Add the value to the variable.

`"_-=`

Subtract the value from the variable.

`"*=", "x=`

Multiply the variable by the value.

`"/=", "÷=`

Divide the variable by the value. If value is 0, does nothing (logs an error).

`"//=`

Integer (floor) division of the variable by the value. If the value is 0, does nothing (logs an error).

`"%=`

Modulo the variable by the value. If the value is 0, does nothing (logs an error).

"^=", "=", "pow"**

Raise the variable to the power of the value.

String Operations

"strip_edges"

Has value: strip leading/trailing whitespace from the value, assign to the variable.

No value: strip leading/trailing whitespace from the variable.

"to_upper"

Has value: convert the value to uppercase, assign to the variable.

No value: convert the variable to uppercase.

"to_lower"

Has value: convert the value to lowercase, assign to the variable.

No value: convert the variable to lowercase.

"replace"

The value must be a 2-element array [search, replacement]. Replace all instances of search with replacement in the variable (returns the variable unchanged if the value isn't a valid array).

"substr"

The value must be a 2-element array [start, length]. Return the substring of the variable starting at start for length characters (returns the variable unchanged if the value isn't a valid array).

"split"

If the variable is a string and the value is a string, split the variable by the value (returns a Godot PackedStringArray-like split result). Otherwise returns the variable unchanged.

"split_to_array"

Same as split, but explicitly converts the result to an Array.

"join"

If the variable is an array and the value is a string, join the array elements using the value as the separator.

"path_join"

If the variable and the value are both strings, join them as a file path.

Array Operations

"append"

If the variable is an array, add the value to the end (does not mutate original array — works on a duplicate).

"append_first"

If the variable is an array, insert the value at the beginning.

"insert"

The value must be a 2-element array [index, item]. Insert item into the variable at index.

"erase"

If the variable is an array, remove the first occurrence of the value from it.

"remove_at"

If the variable is an array and the value is a valid integer index, remove the item at that index.

"clear"

Has value: if the value is an array, return an empty array; if the value is a dictionary, return an empty dictionary; otherwise return the variable unchanged.

No value: same logic applied to the variable instead.

"slice"

The value must be a 2-element array [start, end].

Return the slice of the variable from start to end.

"reverse"

Has value: if the value is an array, return a reversed copy of it.

No value: if the variable is an array, return a reversed copy of it.

"sort"

Has value: if the value is an array, return a sorted copy of it.

No value: if the variable is an array, return a sorted copy of it.

"shuffle"

Has value: if the value is an array, return a shuffled copy of it.

No value: if the variable is an array, return a shuffled copy of it.

"unique"

Has value: if the value is an array, return a copy with duplicate elements removed.

No value: same, applied to the variable.

Dictionary Operations

"merge"

If the variable and the value are both dictionaries, merge the value's keys into a copy of the variable (the value's keys overwrite the variable's on conflict).

"get"

The value must be either a string (key to look up in the variable) or a 2-element array [key, default_value]. Returns the corresponding value from the the variable dictionary, or the default if given and key is missing. Only works if the variable is a dictionary.

"set_key"

The value must be a 2-element array [key, val]. Set key to val in a copy of the variable (the variable must be a dictionary).

"duplicate"

Returns a shallow copy of the value if the value is an array or dictionary; otherwise returns the value unchanged.

"deep_duplicate"

Returns a deep copy of the value if the value is an array or dictionary; otherwise returns the value unchanged.

Type & Conversion

"typeof"

Has value: return the Godot type ID (integer) of the value.

No value: return the type ID of the variable.

"type_string"

Has value: return the human-readable type name of the value.

No value: return the human-readable type name of the variable.

"is_array"

Has value: return true if the value is an array.

No value: return true if the variable is an array.

"is_dict"

Has value: return true if the value is a dictionary.

No value: return true if the variable is a dictionary.

"is_object"

Has value: return true if the value is an Object.

No value: return true if the variable is an Object.

"is_nan"

Has value: return true if the value is NaN (as a float).

No value: return true if the variable is NaN.

"is_inf"

Has value: return true if the value is infinite (as a float).

No value: return true if the variable is infinite.

"to_int"

Has value: convert the value to an integer.

No value: convert the variable to an integer.

"to_float"

Has value: convert the value to a float.

No value: convert the variable to a float.

"to_string"

Has value: convert the value to a string.

No value: convert the variable to a string.

"to_bool"

Has value: convert the value to a boolean.

No value: convert the variable to a boolean.

"toggle_bool"

Has value: return the boolean negation of the value.

No value: return the boolean negation of the variable.

"var_to_str"

Has value: serialize the value to its GDScript string representation.

No value: serialize the variable.

"str_to_var"

Has value: parse the value (converted to string) back into a GDScript variant.

No value: parse the variable the same way.

"json_to_str"

Has value: convert the value to a JSON string.

No value: convert the variable to a JSON string.

"str_to_json"

Has value: parse the value (converted to string) as JSON.

No value: parse the variable as JSON.

"var_to_bytes"

Has value: serialize the value to a PackedByteArray.

No value: serialize the variable.

"bytes_to_var"

If value is a PackedByteArray, deserialize it back into a variant. Otherwise does nothing (logs an error) and returns the variable.

Math Functions

"abs"

Has value: absolute the value of the value.

No value: absolute the value of the variable.

"floor"

Has value: floor of the value.

No value: floor of the variable.

"ceil"

Has value: ceiling of the value.

No value: ceiling of the variable.

"round"

Has value: round the value to nearest integer.

No value: round the variable to nearest integer.

"sqrt", "root"

Has value: square root of the value. If negative, does nothing (logs an error).

No value: square root of the variable. If negative, does nothing (logs an error).

"sign"

Has value: sign of the value (-1, 0, or 1).

No value: sign of the variable.

"sin"

Has value: sine of the value.

No value: sine of the variable.

"cos"

Has value: cosine of the value.

No value: cosine of the variable.

"tan"

Has value: tangent of the value.

No value: tangent of the variable.

"asin"

Has value: arcsine of the value.

No value: arcsine of the variable.

"acos"

Has value: arccosine of the value.

No value: arccosine of the variable.

"atan"

Has value: arctangent of the value.

No value: arctangent of the variable.

"deg_to_rad"

Has value: convert the value from degrees to radians.

No value: convert the variable from degrees to radians.

"rad_to_deg"

Has value: convert the value from radians to degrees.

No value: convert the variable from radians to degrees.

"log"

Has value: natural logarithm of the value. If the value ≤ 0 , does nothing (logs an error).

No value: natural logarithm of the variable. If the variable ≤ 0 , does nothing (logs an error).

"log10"

Has value: base-10 logarithm of the value. If the value ≤ 0 , does nothing (logs an error).

No value: base-10 logarithm of the variable. If the variable ≤ 0 , does nothing (logs an error).

"log_base"

If value is a number and not equal to 1, return the logarithm of the variable in base the value. If the variable ≤ 0 , does nothing (logs an error).

"clamp"

The value must be a 2-element array [min, max]. Clamp the variable between min and max.

"min"

If the value is a non-empty array, return the minimum element of that array. Otherwise return the smaller of the variable and the value.

"max"

If the value is a non-empty array, return the maximum element of that array. Otherwise return the larger of the variable and the value.

"lerp"

The value must be a 2-element array [to, weight]. Linearly interpolate from the variable to "to" by "weight".

"wrap"

The value must be a 2-element array [min, max]. Wrap the variable between min and max.

Random

"randi", "rand_i"

Has value (as a 2-element array): treated as a range — returns a random integer in that range (Deprecated: use "randi_range" instead).

Has value (not an array): returns the variable unchanged.

No value: return a random integer (full int range).

"randi_range", "rand_i_range"

Has value: if it's a 2-element array, return a random integer in that range.

No value: if the variable is a 2-element array, return a random integer in that range.

"randf", "rand_f"

Has value (as a 2-element array): treated as a range — returns a random float in that range (Deprecated: use "randf_range" instead).

Has value (not an array): returns the variable unchanged.

No value: return a random float between 0 and 1.

"randf_range", "rand_f_range"

Has value: if it's a 2-element array, return a random float in that range.

No value: if the variable is a 2-element array, return a random float in that range.

"rand_pick"

Has value: if the value is a non-empty array, return a random element from it.

No value: if the variable is a non-empty array, return a random element from it.

Nodes

"get_node"

If the value is a string, return the node at that path (relative to this node).

"get_node_or_null"

If the value is a string, return the node at that path, or null if it doesn't exist.

